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A Comprehensive AI-Model for H_2-Brine Interfacial Tension Prediction: Implications for Underground Hydrogen Storage

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Abstract

The global focus on clean energy has made hydrogen a crucial part of the future energy framework. Among various large-scale hydrogen technologies, underground hydrogen storage (UHS) in geological formations, particularly saline aquifers, presents a strategic solution. However, the effective storage of hydrogen underground is influenced by fluid-fluid interactions, notably interfacial tension (IFT) between hydrogen and brine. Traditionally, IFT has been measured using experimental methods, such as the pendant drop technique, which are often limited by high costs and technical challenges. This study aims to develop accurate, data-driven machine learning (ML) models to predict H_2–brine IFT under diverse operational conditions, incorporating factors such as temperature, pressure, salinity, and gas composition. To support this, a dataset of IFT measurements was compiled from various literature, covering a wide range of thermodynamic and chemical environments. The dataset included different gas compositions like H_2, CO_2, CH_4, and multiple salt types including NaCl, $CaCl_2$, and KCl. Data preprocessing included outlier removal, unit consistency checks, and Min–Max scaling. ML models using Random Forest (RF) and Artificial Neural Networks (ANNs) were developed due to their strengths in interpretability and pattern recognition. The dataset was split into training, testing, and validation sets for robust evaluation. A strategy was also implemented to generalize input parameters, simplifying the model while maintaining its predictive power. The results showed high prediction accuracy, with the RF model achieving a Mean Absolute Percentage Error (MAPE) of 1.88% on the training data and 3.42% on the testing data. The ANN model achieved MAPE values of 2.14% for training and 3.78% for testing. Temperature ($R = -0.60$) and CO_2 concentration ($R = -0.47$) had the strongest correlations with IFT, while divalent salts like $CaCl_2$ significantly influenced IFT ($R \approx 0.32$–0.33). Generalized input transformations, such as NaCl-equivalent salinity and bulk gas properties, provided stronger correlations with IFT. These findings highlight the innovation of using physically meaningful input generalizations in ML-based IFT modeling, which promotes safer and more efficient underground hydrogen storage systems and supports the transition to a sustainable hydrogen economy.

Introduction

The global transition toward low-carbon energy has brought hydrogen (H_2) to the forefront as a clean, adaptable, and sustainable energy carrier (M and G 2023, Cheekatamarla 2024). Its role in energy distribution and storage has grown in prominence as international efforts intensify to minimize dependence on fossil fuels. Among the various hydrogen storage options, underground hydrogen storage (UHS) in geological formations, particularly in saline aquifers, has emerged as a highly promising solution for large-scale, long-term energy storage (Jafari Raad et al. 2022, Bai et al. 2014, Aneke and Wang 2016, Tarkowski 2019, Cheekatamarla 2024). This approach leverages the unique characteristics of deep saline formations, allowing for effective buffering between renewable energy supply and fluctuating demand. Saline aquifers, due to their wide availability and proven storage potential, have garnered increasing interest for hydrogen storage applications, driven by both technical practicality and economic benefits. However, the success of UHS relies heavily on understanding the complex physical and chemical interactions that occur underground, especially between injected hydrogen and the native brine. In particular, the interfacial tension (IFT) between hydrogen and brine plays a central role in influencing critical processes such as injectivity, capillary trapping, and long-term containment.

The interaction between hydrogen and brine engenders complex flow behaviors that require precise capture to optimize storage. Essential factors such as temperature, pressure, salinity, and gas impurities significantly influence interfacial tension (IFT) and overall hydrogen storage capacity within subsurface environments. Chow et al. (2018) executed IFT experiments on the hydrogen–deionized water system across pressures ranging from 0.5 to 45 MPa and temperatures from 298.15 to 448.15 K, while also evaluating the influence of CO_2 on interfacial behavior. Janjua et al. (2024) expanded upon these findings in H2-brine systems, indicating a reduction in IFT with increasing temperature and a slight decrease with pressure, while heightened salinity resulted in increased IFT. Al-Mukainah et al. (2022) noted a marginal pressure-dependent decrease in IFT. Conversely, Mirchi et al. (2022) discovered that escalating temperature or methane (CH4) concentration in the gas phase further diminished H2–methane (CH4)/brine IFT. Higgs et al. (2022) investigated the effects of salinity and pressure, observing a reduction in IFT with pressure, yet found no discernible correlation between salinity, particularly sodium chloride (NaCl), and IFT.

Historically, interfacial tension (IFT) has been measured using the pendant drop technique under controlled laboratory conditions. However, hydrogen's high volatility and the complexity of replicating subsurface environments make direct experimentation difficult. Such lab tests are often time-consuming, costly, and prone to variability, especially under high-pressure, high-temperature (HPHT) scenarios. Machine learning (ML) provides a data-driven approach, enabling the capture of complex patterns from existing data to deliver fast, scalable, and accurate predictions. Developing robust ML models is therefore crucial for reliable IFT estimation in H_2–brine systems, influencing key aspects of hydrogen storage such as injectivity, capillary behavior, and containment safety. This study contributes to advancing subsurface hydrogen storage technologies—critical for balancing seasonal energy needs, integrating renewables, and supporting the global transition to a low-carbon hydrogen economy. Machine learning has increasingly proven valuable in geoscience and energy applications, with hybrid and ensemble techniques driving notable gains in predictive performance. These methods have shown strong potential in areas such as estimating rock petrophysical properties and modeling carbon capture and CO_2 storage in saline aquifers. Their success underscores ML's expanding influence in improving forecasting precision and informing better operational decision-making across various domains (Al-Mudhafar et al. 2022, Vo Thanh et al. 2022, Zhang et al. 2023, Ibrahim 2022).

Despite progress in experimental and computational methods, accurately predicting interfacial tension (IFT) in H_2–brine systems under realistic subsurface conditions remains difficult. This challenge arises from hydrogen's volatility and varying reservoir factors such as pressure, temperature, salinity, and gas composition. Current models often struggle to generalize across different chemical environments. This study presents machine learning models that account for various influencing factors, including gas impurities like CH_4 and CO_2 and different types of salts. A key innovation is the application of physically meaningful input transformations, such as salt equivalency and gas property generalization, which streamline input complexity without sacrificing accuracy. This approach improves model interpretability and facilitates dependable IFT predictions for hydrogen storage in saline aquifers.

Methodology

Data Collection and Preparation

A comprehensive and diverse dataset was assembled from multiple published studies, capturing IFT measurements across a wide range of conditions involving pressure, temperature, salinity, and gas composition (e.g., (Chow et al. 2018, Al-Mukainah et al. 2022, Higgs et al. 2022, Mirchi et al. 2022, Alanazi et al. 2023, Janjua et al. 2024). Rigorous data cleaning procedures were implemented to ensure the reliability and relevance of the data. This included unit standardization, removal of outliers and duplicate entries, and thorough consistency checks. Dataset validity was further assessed through visual inspection and statistical diagnostics, such as skewness analysis, to confirm proper data distribution and quality. The dataset comprises approximately 500 entries, with univariate statistics summarized in Table 1. Temperatures range from 71.6°F to 348.3°F, and pressures span from 14.5 psi to 6,554 psi. Gas impurity concentrations, including CO_2 and CH_4, vary from pure hydrogen (0%) to fully substituted mixtures (100%). Total salinity is expressed through molar concentrations of key salts—Na_2SO_4, NaCl, $CaCl_2$, $MgCl_2$, and KCl—each ranging from 0.0 M to 3.0 M. IFT values fall between 24.2 mN/m and 89.0 mN/m. High standard deviations for pressure (1,724 psi) and temperature (51.6°F) indicate broad variability. Skewness analysis shows a positive skew for temperature and most salts, implying right-tailed distributions, while IFT displays slight negative skewness, suggesting a tendency toward lower-end values.

Table 1—Statistical analysis of input and output parameters using univariate methods

	T	P	CO_2	CH_4	$NaCHO_3$	Na_2SO_4	NaCl	$CaCl_2$	$MgCl_2$	KCl	IFT
	F	psi	%	%	M	M	M	M	M	M	mN/m
Minimum	71.6	14.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	30.1
Maximum	348	6554	70.0	100.0	0.0	3.0	3.0	3.0	3.0	3.0	89.0
Skewness coefficient	2.2	1.4	3.0	2.2	5.5	4.8	2.0	4.8	4.8	4.8	-0.6
Mean	126	1532	4.8	11.0	0.0	0.1	0.4	0.1	0.1	0.1	64.3

Figure 1 illustrates the correlation coefficients between interfacial tension (IFT) and input variables, highlighting both the magnitude and direction of their influence. Among the salts, $CaCl_2$ (0.33), Na_2SO_4 (0.32), and $MgCl_2$ (0.32) show strong positive correlations with IFT, suggesting that higher concentrations lead to increased IFT. In contrast, NaCl (0.10) and KCl (0.11) exhibit weaker positive correlations, implying a limited impact. Temperature demonstrates a strong inverse relationship with IFT (–0.60), making it the most significant factor in lowering IFT—likely due to enhanced molecular activity weakening interfacial forces. CO_2 (–0.47) and CH_4 (–0.27) also negatively correlate with IFT, with CO_2 having a greater effect, potentially due to its higher solubility. Pressure shows a modest negative correlation (–0.24), indicating a minor role in reducing IFT. In summary, temperature and CO_2 are the most influential in reducing IFT, while salts like $CaCl_2$ and Na_2SO_4 are key contributors to its increase. To ensure uniform data scaling and improve model training, especially in neural networks, all features were normalized using the Min–Max method.

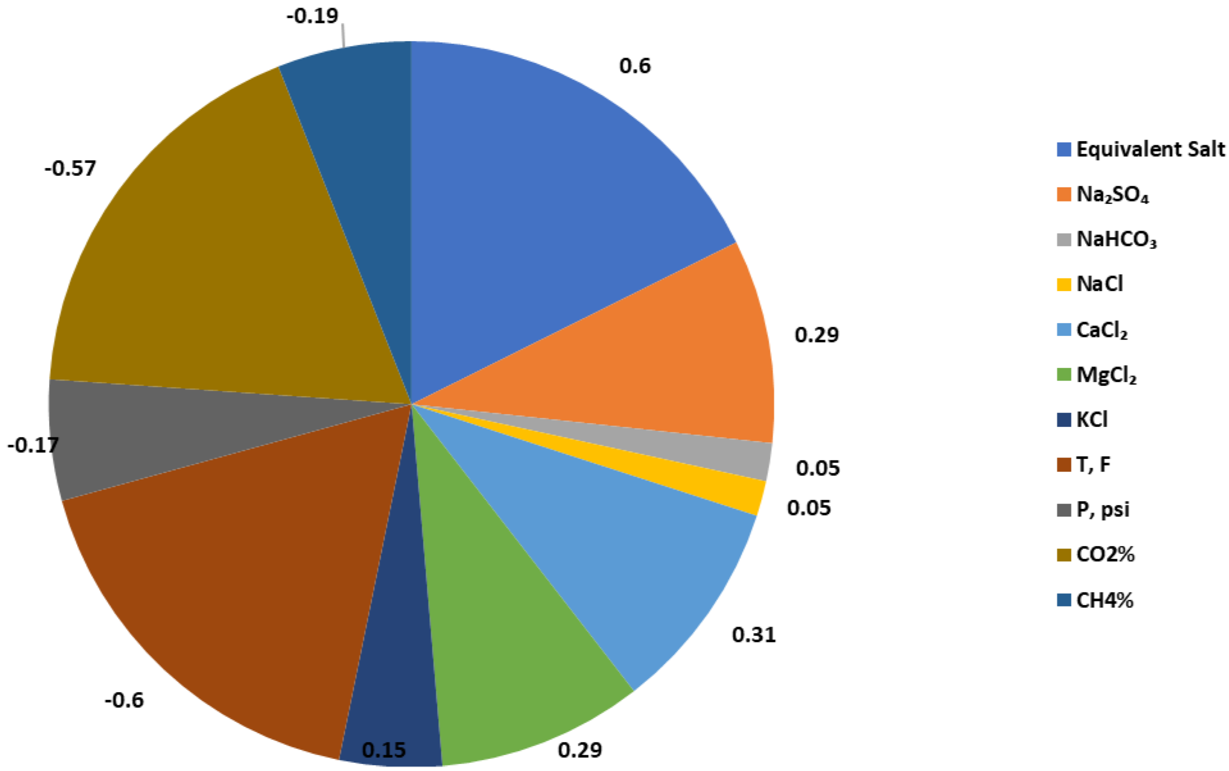

Figure 1—The correlation coefficient between the IFT and various input parameters.

Model Development

Figure 2 presents the structured workflow for developing machine learning models. The process begins with Data Preparation, which involves collecting and preprocessing relevant, high-quality data for model input. A random subsampling cross-validation strategy is employed, where datasets are split into training and testing sets using varying ratios (e.g., 40/60 to 20/80) to evaluate model stability across different data partitions. Next is the Model Training phase, where hyperparameters are optimized to enhance predictive accuracy. Following training, Model Testing is performed, enabling adjustments to the train-test split as needed for performance improvement. The Model Validation phase assesses generalization by testing the model on 100 previously unseen data points, ensuring its reliability on out-of-sample predictions. After validation, a Sensitivity Analysis identifies how each input variable influences the model's output.

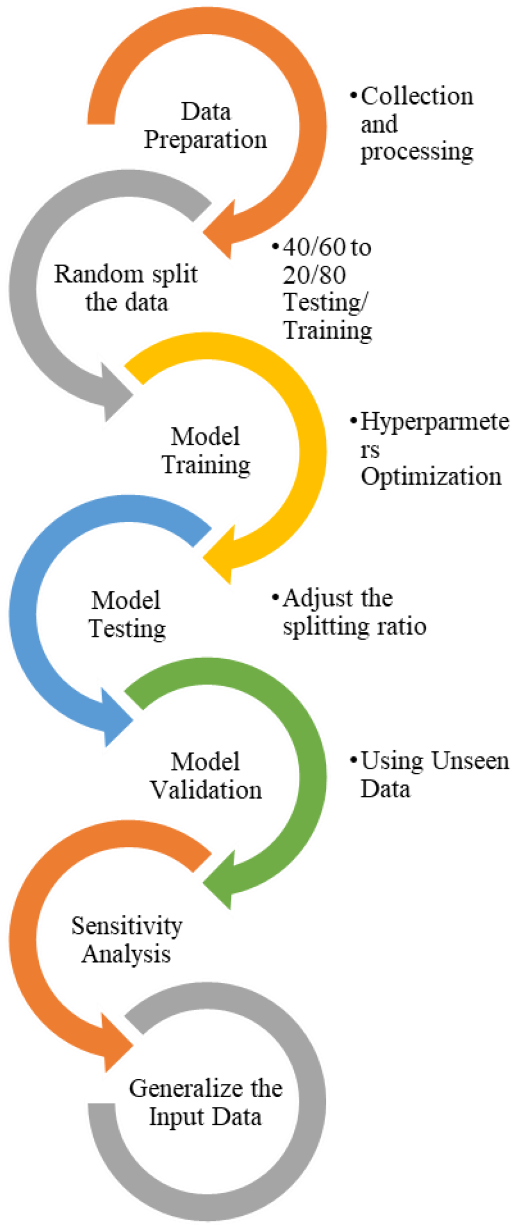

Figure 2—The different Procedures used to develop the different ML Models.

The final step is Input Generalization, where input features are transformed or simplified to reduce model complexity while maintaining broad applicability across diverse input scenarios. Model performance is assessed using metrics such as the coefficient of determination (R^2) and Mean Absolute Percentage Error (MAPE), providing a comprehensive view of prediction accuracy and robustness.

$$R^2 = 1 - \frac{SS_E}{SS_{YY}} \tag{1}$$

$$SS_E = \Sigma_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2 \tag{2}$$

$$S_{YY} = \Sigma_{i=1}^{n}\left(y_i - \bar{y}\right)^2 \tag{3}$$

$$MAPE = \frac{100}{n}\Sigma_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{4}$$

To develop robust and accurate models for predicting hydrogen–brine interfacial tension (IFT), a diversified machine learning (ML) strategy was adopted, incorporating both **Random Forest (RF)** and **Artificial Neural Networks (ANNs)**. These algorithms were selected due to their complementary strengths in modeling nonlinear, multivariate systems common in subsurface energy applications. Together, they offer a balanced evaluation between transparency and predictive depth.

Random Forest (RF): Interpretability and Robustness

Random Forest is an ensemble learning method introduced by Breiman (2001) that is particularly effective for regression and classification tasks involving high-dimensional and noisy datasets as shown in Figure 3. The RF algorithm operates by constructing a large number of independent decision trees, each trained on a different bootstrap sample of the dataset. During tree construction, a random subset of features is selected at each split, ensuring model diversity. The final prediction is obtained by averaging the outputs of all trees (for regression tasks), which stabilizes predictions and significantly reduces variance compared to single decision trees.

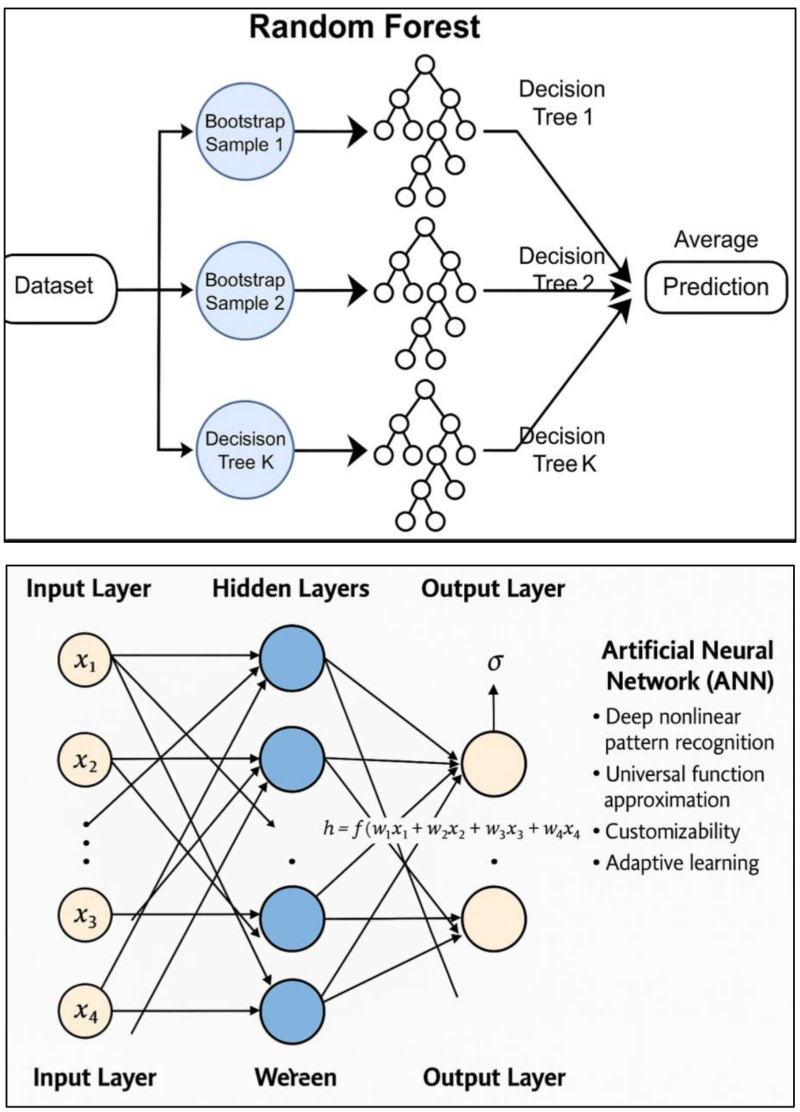

Figure 3—Architecture of Random Forest (left) and Artificial Neural Network (right) models used for IFT prediction, illustrating their data flow and structural components.

Key advantages of RF in this context include:

- **Nonlinear modeling capability**: RF naturally captures interactions between variables such as temperature, salinity, gas impurities, and pressure.
- **Resistance to overfitting**: Due to averaging over many decorrelated trees, RF models are less prone to overfitting, especially when the dataset contains noise or redundant features.
- **Feature importance estimation**: RF provides a ranked output of variable importance, offering insights into which input parameters most significantly affect IFT.
- **Scalability and robustness**: RF handles both small and large datasets effectively, with minimal need for hyperparameter tuning.

Artificial Neural Networks (ANN): Flexibility and Depth

Artificial Neural Networks (ANNs) are inspired by the structure of the human brain, consisting of interconnected nodes (neurons) organized into input, hidden, and output layers as shown in Figure 3. Each connection has an associated weight, and each neuron applies a nonlinear activation function to its inputs, allowing the network to model complex, highly nonlinear relationships.

In this study, ANNs are particularly suitable due to the following properties:

- **Deep nonlinear pattern recognition**: ANNs can capture complex dependencies between input variables (e.g., pressure, temperature, impurity concentrations, and salt type) that traditional models may fail to identify.
- **Universal function approximation**: Given sufficient hidden layers and neurons, ANNs can approximate any continuous function to an arbitrary degree of accuracy, making them ideal for modeling subtle IFT variations across diverse reservoir conditions.
- **Customizability**: ANNs can be tailored in terms of architecture (number of layers, neurons per layer), activation functions (ReLU, tanh, sigmoid), and optimization techniques (Adam, SGD), allowing flexibility in adapting to specific modeling needs.
- **Adaptive learning**: With backpropagation and gradient descent, ANNs iteratively minimize prediction error, learning complex interactions even when explicit physical relationships are unknown.

However, ANNs are often regarded as "black-box" models due to limited interpretability—one of the reasons RF is used alongside to provide transparent insight.

Modeling Strategy and Synergy

By combining RF and ANNs, the modeling framework capitalizes on their respective strengths:

- RF supports interpretability, fast training, and a built-in mechanism for understanding variable importance.
- ANN offers unmatched flexibility and predictive power for capturing intricate relationships in the data.

This dual approach not only improves predictive accuracy but also enhances the credibility and generalizability of the findings. It ensures that the developed models are suitable for real-world underground hydrogen storage (UHS) applications, where physical processes are both multivariate and nonlinear, and data availability may vary across regions and conditions.

Results and Discussion

Figure 4 presents cross-plots of actual versus predicted IFT values for both training and test datasets:

- The RF model achieved MAPE values of 1.88% for training and 3.42% for testing, reflecting high prediction accuracy and excellent generalization with minimal overfitting.
- The ANN model reached a training MAPE of 2.14% and testing MAPE of 3.78%, indicating strong predictive capability, though slightly less consistent than RF on unseen data.

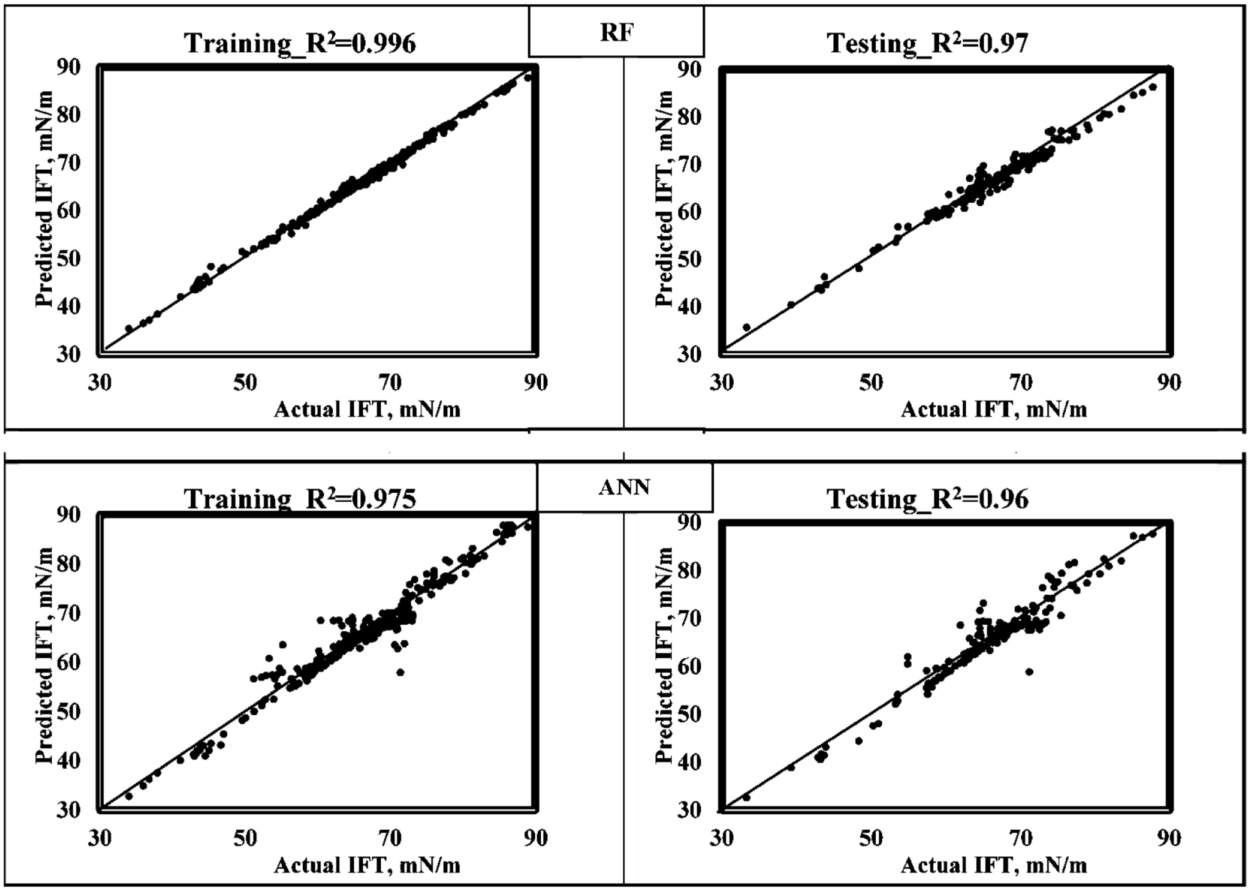

Figure 4—Cross plots for the actual versus the predicted IFT values from different ML models for both training and testing datasets.

In both cases, data points closely cluster around the 45-degree reference line, confirming the models' effectiveness in approximating real IFT values.

The comparative analysis reveals that:

- Random Forest delivers highly accurate and stable predictions with added interpretability, making it well-suited for scenarios where understanding feature influence is critical (e.g., the role of specific salts or temperature).
- Artificial Neural Networks offer flexible modeling of deeply nonlinear behaviors and perform well across varied input conditions, although their performance may benefit from additional hyperparameter tuning or regularization to reduce residual variance.

In conclusion, both RF and ANN exhibit strong potential for practical deployment in modeling interfacial behavior in underground hydrogen storage, with MAPE-based evaluation confirming their reliability and real-world applicability.

Figure 5 presents the correlation coefficients between interfacial tension (IFT) and each input parameter, providing insight into the linear relationships within the dataset. This analysis offers a preliminary understanding of how each factor independently influences IFT values between hydrogen and brine.

Among all input variables, temperature exhibits the strongest negative correlation (R ≈ –0.60), indicating that as temperature increases, IFT consistently decreases. This trend likely results from enhanced molecular agitation at higher temperatures, which disrupts cohesive forces at the gas–liquid interface. CO_2 concentration also shows a significant negative correlation (R ≈ –0.47), reflecting its high solubility and its tendency to reduce interfacial tension by altering the chemical environment. CH_4, while less impactful, follows the same direction with a weaker negative correlation (R ≈ –0.27). Operating pressure contributes to a lesser extent (R ≈ –0.24), but still indicates a mild tendency to reduce IFT. In contrast, most salts demonstrate positive correlations, suggesting that increasing salinity generally raises IFT. Notably, divalent salts like $CaCl_2$ (R ≈ 0.33) and $MgCl_2$ (R ≈ 0.32) show stronger effects compared to monovalent salts such as NaCl (R ≈ 0.10) and KCl (R ≈ 0.11). Na_2SO_4, although technically a sodium salt, also exhibits a moderately positive correlation (R ≈ 0.32) due to the influence of the sulfate ion. These correlation patterns affirm that temperature and CO_2 concentration play major roles in reducing IFT, while certain salt types, particularly divalent ions, tend to increase IFT. Although this analysis only captures linear relationships, it provides a valuable first look at variable influence prior to more detailed nonlinear modeling.

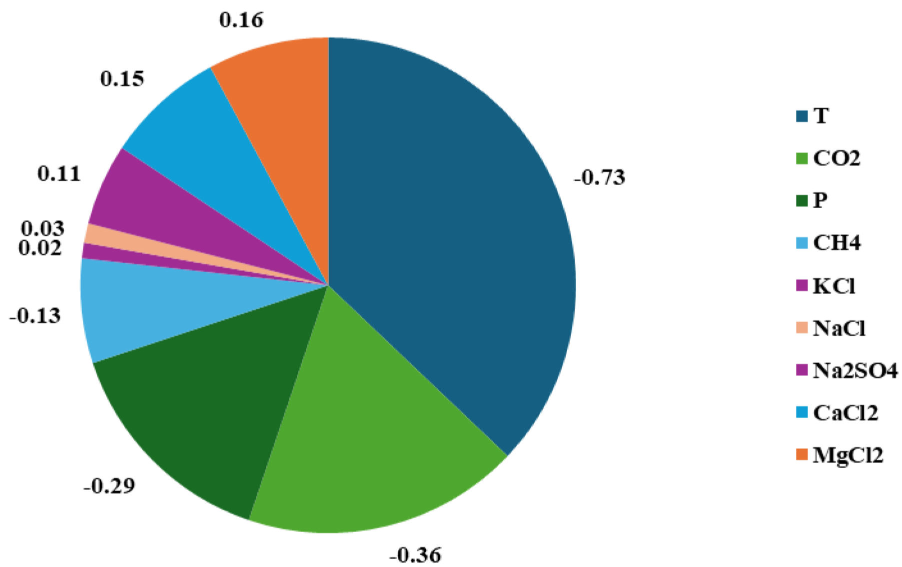

Figure 5—Correlation coefficient for input parameters versus IFT, highlighting their importance based on sensitivity analysis.

Generalize Input Parameters

To enhance model efficiency and reduce complexity, input parameters were consolidated into equivalent representations that preserve their physical impact. This generalization aimed to capture the cumulative influence of related variables while reducing dimensionality.

For salinity, all individual salts were converted into a unified NaCl-equivalent concentration, calculated using the following equation:

$$NaClEq. = \Sigma_1^n Salt\text{concentration} * \frac{Mw_{salt}}{Mw_{NaCl}} \tag{5}$$

For gas composition, simplification was achieved by introducing bulk parameters that reflect the collective properties of gas mixtures: specific gas gravity, Pseudo-critical pressure, and Pseudo-Critical temperature. These generalized variables outperformed individual gas component concentrations in explaining IFT variability. Correlation coefficients in Figure 6 improved to –0.56 (γ_9), –0.49 (Ppc), and – 0.48 (Tpc), compared to values below –0.47 for individual gases such as CO_2 and CH_4. This input transformation strategy not only enhances model interpretability but also strengthens the correlation with IFT, making the model more robust and generalizable across various hydrogen storage scenarios.

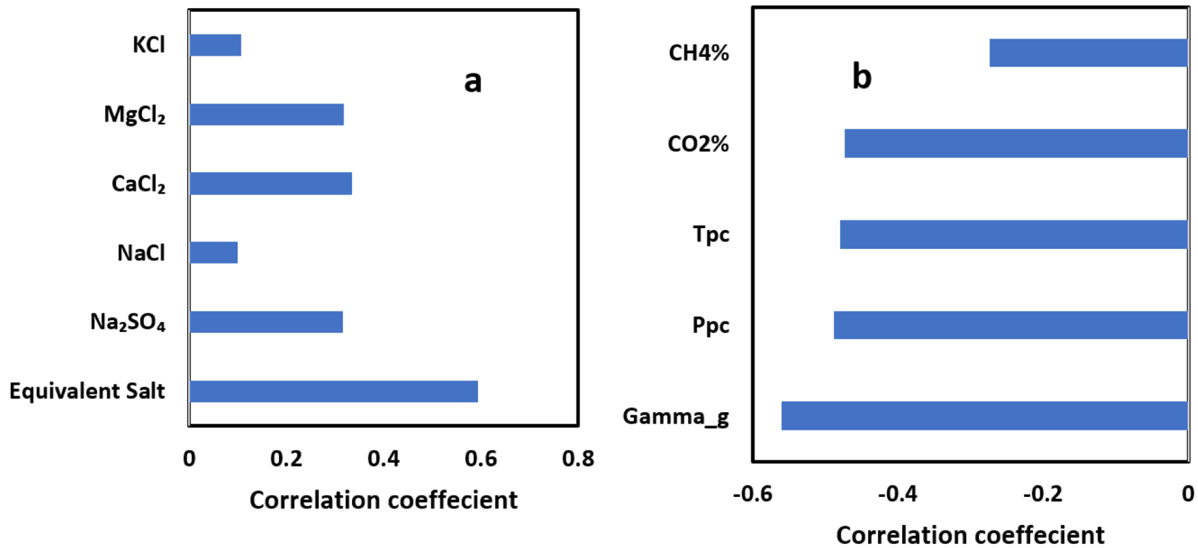

Figure 6—Correlation the coefficients associated with the interfacial tension (IFT) and various input parameters are as follows: a) the equivalency of salt in relation to the individual composition of salts, and b) a comparison of gas equivalent parameters with respect to the gas composition.

Figure 7, which compares Random Forest (RF) model predictions using both the full set of original input parameters and their generalized equivalents, demonstrates a critical advancement in model simplification and generalization. This transformation, which involves converting individual salt types into NaCl-equivalent concentration and summarizing gas composition using bulk gas properties (e.g., specific gravity, pseudo-critical pressure, and temperature), has multiple significant benefits for both modeling efficiency and practical deployment.

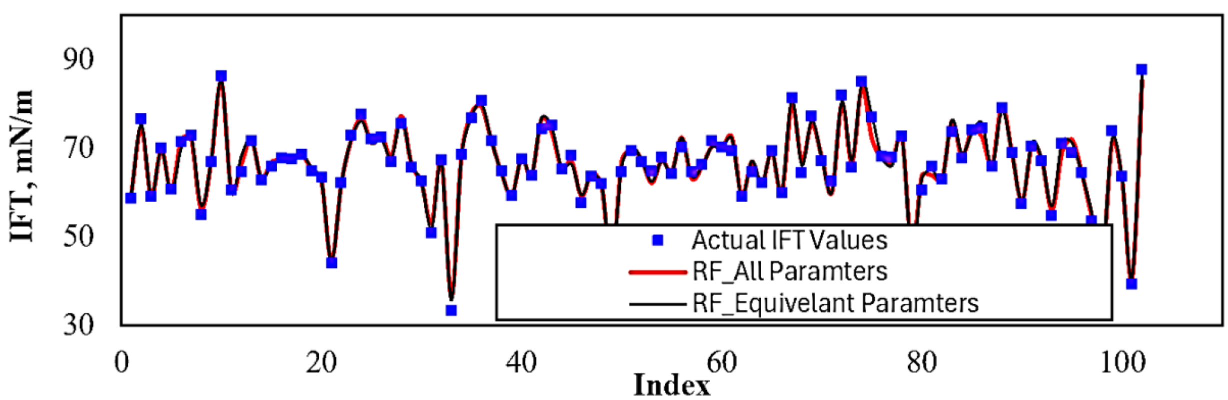

Figure 7—RF model results for IFT prediction based on the equivalent parameters on validation data sets.

Dimensionality Reduction Without Performance Loss

The close alignment between actual IFT values and predictions from both the full-input and equivalent-parameter models confirms that generalized inputs retain the predictive power of the original high-dimensional data. This shows that essential physical trends—such as how salinity and gas composition affect IFT—can be effectively captured with fewer variables. Reducing the number of input parameters:

- Minimizes overfitting risks, especially with limited datasets.
- Speeds up model training and inference time.
- Makes the model easier to update or retrain with new data.

Enhanced Interpretability and Physical Meaning

Generalized parameters are grounded in physical chemistry—for example, NaCl-equivalent accounts for molecular weight differences across salts, and pseudo-critical properties reflect thermodynamic gas behavior. These physically meaningful transformations allow scientists and engineers to interpret model behavior more intuitively:

- The NaCl-equivalent simplifies salinity into a single continuous variable that reflects both concentration and salt type.
- γ_9, Ppc, and Tpc serve as proxies for gas effects, avoiding the need to model each impurity separately.

This clarity supports better decision-making in reservoir engineering, particularly in evaluating which environmental factors most influence IFT during hydrogen storage.

Improved Correlation and Predictive Consistency

The transformation significantly boosted correlation coefficients between inputs and IFT:

- NaCl-equivalent improved correlation to $R \approx 0.60$, compared to $R = 0.10$–0.33 for individual salts.
- Bulk gas properties (γ_9, Ppc, Tpc) yielded $R = -0.56$ to -0.48, outperforming individual gases ($R < -0.47$).

These enhancements result in more stable and consistent predictions, particularly in scenarios where the exact composition of the gas or the specific salt mixture may vary. The generalization thus supports robust forecasting in real-world settings with uncertainty or incomplete compositional data.

Practical Value for Underground Hydrogen Storage (UHS)

In underground hydrogen storage projects, subsurface conditions vary across formations and operational phases. Generalized inputs:

- Allow the same model to be applied across different reservoirs, brine chemistries, and gas injection compositions.
- Enable deployment of fewer, more adaptable models, which is valuable for real-time monitoring, feasibility studies, and risk assessments.

The use of generalized input parameters in ML modeling of IFT is not just a computational convenience—it is a strategic enhancement that improves the model's accuracy, efficiency, and transferability. This methodology bridges the gap between data-driven modeling and physical interpretability, making it a powerful tool for advancing hydrogen storage technology in saline aquifers.

Conclusions

This study focused on developing accurate and generalizable machine learning (ML) models to predict the interfacial tension (IFT) between hydrogen and brine under varying subsurface conditions, with direct implications for underground hydrogen storage (UHS) in saline aquifers. The models incorporated diverse chemical and thermodynamic variables, introducing generalized input transformations to enhance scalability and interpretability. These are the main findings:

- A curated dataset of approximately 1,000 IFT measurements was compiled from published literature, spanning wide ranges of temperature (71.6–348.3°F), pressure (14.5–6554 psi), gas compositions (0–100% CH_4 and CO_2), and salinity profiles using five different salts.

- Random Forest (RF) and Artificial Neural Networks (ANNs) were identified as the top-performing models, achieving MAPE values of 1.88% (RF training), 3.42% (RF testing), and 3.78% (ANN testing), with R^2 values exceeding 0.97, demonstrating excellent predictive performance.
- Temperature and CO_2 concentration were the most significant factors in lowering IFT, with correlation coefficients of –0.60 and –0.47, respectively, while $CaCl_2$ and $MgCl_2$ showed the strongest salinity-related increases in IFT ($R \approx 0.32$–0.33).
- A NaCl-equivalent salinity index was developed to unify the effects of multiple salts into a single metric, increasing the correlation with IFT from R = 0.10–0.33 (individual salts) to R = 0.60.
- For gas composition, the use of bulk properties—such as specific gravity (γg), pseudo-critical pressure (Ppc), and pseudo-critical temperature (Tpc)—enhanced correlation with IFT (R = –0.56 to –0.48), outperforming models that used individual gas concentrations.
- Validation results using 100 unseen data points confirmed that the generalized-input RF model maintained predictive accuracy comparable to the full-input model, indicating strong generalizability across different H_2 storage conditions.

These findings demonstrate the novelty and utility of applying physically informed generalization techniques in ML-based IFT prediction, enabling accurate, interpretable, and broadly applicable modeling for hydrogen storage in saline aquifers.

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